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A COMPREHENSIVE FINANCIAL PERFORMANCE ANALYSIS OF AMUL (GCMMF): A STUDY OF PROFITABILITY, LIQUIDITY, AND SOLVENCY TRENDS

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ABSTRACT:

The Gujarat Cooperative Milk Marketing Federation (GCMMF), operating under the brand identity of Amul, stands as India's largest food products cooperative and a defining institution of the country's dairy economy. This study undertakes a systematic financial performance analysis of GCMMF spanning fiscal years 2018–19 to 2022–23, examining profitability, liquidity, and solvency dimensions to assess organisational health and strategic trajectory. Employing ratio analysis as the principal methodological instrument — encompassing return on equity, net profit margin, current ratio, quick ratio, debt-to-equity ratio, and interest coverage ratio

— the investigation draws upon five consecutive years of audited annual reports. Findings reveal a consistent upward revenue trajectory peaking at INR 72,000 crore in FY 2022–23, while net profit margins remain structurally compressed within the 2–3% band, reflecting cooperative pricing philosophy and inflationary input cost pressures. Liquidity analysis indicates that current ratios consistently exceed the benchmark threshold of 1.5, affirming short-term obligation capacity. Solvency indicators demonstrate a conservative financial structure with low leverage, evidencing limited dependence on external debt financing. These findings collectively suggest that Amul's financial architecture prioritises farmer welfare and volume-driven growth over margin maximisation — a deliberate strategic posture that warrants ongoing evaluation in the context of intensifying competition from private dairy multinationals. The study contributes actionable intelligence for cooperative governance, policy formulation, and academic understanding of social enterprise finance in emerging markets.

Keywords: Financial Ratio Analysis; Amul; GCMMF; Profitability; Liquidity; Solvency; Cooperative Finance; Dairy Sector

Introduction

1.1 Amul and the Indian Cooperative Dairy Architecture

Amul, the consumer-facing brand of the Gujarat Cooperative Milk Marketing Federation (GCMMF), represents one of the most consequential economic experiments in post-independence India. Established in 1946 under the visionary stewardship of Tribhuvan Das Patel and subsequently developed by Dr. Verghese Kurien, Amul served as the institutional blue print for Operation Flood — the world's largest dairy development programme — which transformed India from a milk-deficient nation into the single largest producer of milk globally (Kurien, 2005). The cooperative model underlying Amul is structurally distinctive: farmers at the village level own primary cooperatives, which aggregate into district unions, which in turn consolidate under GCMMF for national marketing and distribution. This three-tier architecture ensures that economic surplus circulates back to over 3.6 million dairy farmer members rather than accruing to private shareholders (GCMMF Annual Report, 2023).

From a financial performance standpoint, Amul occupies an unusual position. Unlike conventional for-profit corporations wherein margin maximization constitutes the primary managerial objective, GCMMF's financial strategy is explicitly subordinated to its social mandate of farmer income enhancement. This orientation produces a distinctive financial profile — high revenue volumes, structurally compressed margins, and conservative capital structures — that demands analytical frameworks sensitive to cooperative enterprise logic rather than conventional commercial benchmarks (Shah, 2016). Understanding this financial profile is not merely an academic exercise; it carries direct implications for regulatory oversight, cooperative governance reform, competitive strategy, and rural development policy.

1.2 Rationale and Significance of the Study

The Indian dairy sector has experienced profound structural transformation over the past decade. Private sector participation has intensified markedly, with multinationals such as Nestlé, Danone, and Lactalis expanding their Indian operations alongside domestic private players including Heritage Foods and Parag Milk Foods. This competitive intensification raises pertinent questions regarding Amul's financial resilience: Can the cooperative maintain its revenue growth trajectory while sustaining farmer-friendly procurement prices? Are its liquidity buffers adequate to absorb demand shocks? Does its solvency structure offer sufficient protection against systemic economic disruptions?

Existing scholarly literature on Amul is predominantly qualitative, focusing on institutional design, governance mechanisms, and rural development outcomes (Banerjee, Mookherjee, Munshi, & Ray, 2001). Systematic, multi-year quantitative financial analysis employing standardised ratio methodologies remains comparatively underdeveloped. This study addresses that gap through a structured five-year financial performance investigation covering FY 2018–19 through FY2022–23, a period encompassing pre-pandemic stability, COVID-19 disruption, post-pandemic recovery, and inflationary pressure — offering a comprehensive stress-test of Amul's financial architecture.

1.3 Research Objectives

The study pursues four interconnected research objectives:

- (i) To examine profitability trends across five fiscal years using return on equity, net profit margin, and gross profit margin ratios.
- (ii) To evaluate short-term liquidity adequacy through current ratio and quick ratio analysis.
- (iii) To assess long-term solvency health via debt-to-equity ratio and interest coverage ratio.

To synthesise findings in to strategic and policy recommendations aligned with cooperative enterprise principles.

Literature Review with Research Gap

2.1 Financial Performance Analysis in Cooperative Enterprises

The theoretical literature on cooperative financial performance diverges substantially from conventional corporate finance paradigms. Hendrikse and Veerman (2001) posit that agricultural cooperatives face inherent tension between their dual roles as farmer-owned service organisations and market-competing commercial entities. This structural duality produces what Chaddad and Cook (2004) term the 'vaguely defined property rights problem,' wherein diffused ownership dilutes accountability for financial efficiency, potentially manifesting in suboptimal capital allocation. However, Bijman et al. (2012) counter that cooperative enterprises with strong governance frameworks can achieve financial performance metrics comparable to investor-owned firms while delivering superior distributional outcomes to member-farmers.

In the Indian dairy cooperative context, Hanumantha Rao (2001) documented that GCMMF consistently outperformed state dairy development boards on operational efficiency metrics while simultaneously delivering higher milk procurement prices to farmers — suggesting that cooperative mission and financial performance need not be structurally antagonistic. More recently, Mishra and Kumar (2019) analysed the financial statements of selected Indian dairy cooperatives and found that leverage ratios remained significantly below industry averages, attributed to preferential government credit access and member equity infusions, a finding with direct relevance to the present investigation.

2.2 Ratio Analysis as Financial Performance Methodology

Ratio analysis constitutes one of the most widely employed methodological instruments in corporate financial evaluation, offering standardised metrics that facilitate both longitudinal trend analysis and cross-sectional benchmarking (Brigham & Houston, 2019). The methodological literature distinguishes among profitability ratios — which assess value generation capacity — liquidity ratios — which evaluate short-term obligation management — and solvency or leverage ratios — which gauge long-term financial stability. Pandey (2015) emphasises that ratio analysis achieves its highest diagnostic utility when applied longitudinally across multiple accounting periods, permitting identification of structural trends that single-period snapshots obscure. This principle directly informs the five-year analytical framework adopted in the present study.

Ross, Westerfield, and Jordan (2019) caution, however, that ratio analysis possesses inherent limitations: ratios are derived from historical cost accounting data that may not reflect current market values; they are sensitive to accounting policy choices such as depreciation methods and inventory valuation approaches; and they do not capture qualitative dimensions including brand equity, governance quality, and relational capital — all of which are particularly significant in cooperative enterprises. These limitations are acknowledged in the present study's interpretive framework.

2.3 Identified Research Gap

A systematic review of extant literature reveals three notable gaps that the present study addresses. First, while Amul has been extensively studied through qualitative institutional lenses, rigorous multi-year quantitative financial ratio analysis employing a comprehensive suite of profitability, liquidity, and solvency metrics is absent from the scholarly record. Second, the specific five-year period spanning FY 2018–19 to FY 2022–23 — encompassing COVID-19 disruption, subsequent recovery, and inflationary pressures — has not been subjected to structured financial examination. Third, existing studies rarely contextualise Amul's financial ratios against cooperative enterprise benchmarks, defaulting instead to conventional corporate norms that may not accurately reflect cooperative financial health. The present investigation directly addresses all three identified gaps.

Research Design/Model

3.1 Research Design and Methodological Approach

This study adopts a descriptive and analytical research design grounded in secondary data analysis. The methodological approach is quantitative in orientation, employing financial ratio analysis as its primary analytical instrument. Descriptive research design is appropriate for the study's objectives insofar as it seeks to systematically document and interpret financial performance patterns across a defined temporal period rather than establishing causal relationships or testing predictive models. The longitudinal dimension — spanning five fiscal years — enables trend identification and temporal pattern analysis that cross-sectional designs cannot accommodate.

3.2 Data Source and Collection Procedure

All financial data are sourced exclusively from GCMMF's officially published audited annual reports for fiscal years 2018–19 through 2022–23, supplemented by disclosures on the organisation's official corporate website and the Ministry of Corporate Affairs (MCA) filings where applicable. Annual reports constitute primary financial disclosure documents prepared in accordance with Indian Generally Accepted Accounting Principles (Ind AS), providing standardised, auditor-verified financial statements including balance sheets, profit and loss accounts, and cash flow statements. No primary data collection was undertaken, consistent with the secondary research design.

3.3 Analytical Framework: Financial Ratios Employed

The study employs the following six financial ratios organized across three analytical categories:

Category	Ratio	Formula
Profitability	Net Profit Margin	$\text{Net Profit} / \text{Net Sales} \times 100$
Profitability	Return on Equity (ROE)	$\text{Net Profit} / \text{Shareholders' Equity} \times 100$
Profitability	Gross Profit Margin	$\text{Gross Profit} / \text{Net Sales} \times 100$
Liquidity	Current Ratio	$\text{Current Assets} / \text{Current Liabilities}$
Liquidity	Quick Ratio	$(\text{Current Assets} - \text{Inventory}) / \text{Current Liabilities}$
Solvency	Debt-to-Equity Ratio	$\text{Total Debt} / \text{Shareholders' Equity}$
Solvency	Interest Coverage Ratio	$\text{EBIT} / \text{Interest Expense}$

Table 1: Financial Ratios Employed in the Study

3.4 Analytical Period and Limitations

The analytical period spans five fiscal years (FY 2018–19 to FY 2022–23), selected to encompass a full business cycle including pre-pandemic stability, the COVID-19 shock year (FY 2019–20 and FY 2020–21), and subsequent recovery. This period provides sufficient temporal breadth to distinguish structural financial patterns from transient anomalies. Methodological limitations include: reliance on consolidated GCMMF financials that aggregate district union performance, potentially masking intra-federation variation; use of

Book value equity in ratio computation rather than market value (as GCMMF shares are not publicly traded); and the inherent limitations of ratio analysis noted in the literature review. These limitations are acknowledged throughout the interpretive discussion.

1. Data Analysis and Findings

4.1 Revenue and Profitability Analysis

Amul's revenue performance across the five-year study period demonstrates a robust and uninterrupted growth trajectory. Total turnover expanded from approximately INR 29,220 crore in FY 2018–19 to INR 72,000 crore in FY2022–23, representing a compound annual growth rate (CAGR) of approximately 25.3%. This growth reflects both volume expansion driven by geographic market deepening and product portfolio diversification into value-added categories including cheese, ice cream, paneer, and UHT milk — segments commanding superior price realisation relative to commodity liquid milk.

Fiscal Year	Net Revenue (INR Crore)	Gross Profit Margin (%)	Net Profit Margin (%)	ROE (%)	Net Profit (INR Crore)
2018–19	29,220	7.8	2.1	11.4	613
2019–20	38,550	8.1	2.3	12.7	887
2020–21	39,248	7.6	2.0	11.1	786
2021–22	61,000	8.4	2.5	14.2	1,525

2022–23	72,000	8.7	2.6	15.3	1,872
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Table 2: Revenue and Profitability Indicators –GCMMF (FY 2018–19 to FY 2022–23)

Despite impressive revenue expansion, net profit margins remain persistently compressed within the 2.0–2.6% band across all five years studied. This pattern warrants careful interpretive contextualisation. In conventional commercial enterprises, such thin margins would constitute a performance deficiency warranting managerial corrective action. Within Amul's cooperative framework, however, margin compression is a deliberate outcome of the organisation's pricing philosophy: procurement prices paid to farmer-members are maintained at approximately 75–80% of consumer prices, substantially above commercial dairy processor norms, with this premium constituting the cooperative's primary value distribution mechanism (GCMMF Annual Report, 2023). Accordingly, net profit figures significantly understate the total economic value generated for farmer-members.

Return on equity demonstrated a gradual upward trajectory from 11.4% in FY 2018–19 to 15.3% in FY 2022–23, with a temporary dip to 11.1% in FY 2020–21 corresponding to the COVID-19 disruption year. This recovery trajectory suggests organisational resilience and effective post-pandemic operational adaptation. The FY 2020–21 margin contraction is attributable to elevated raw milk procurement costs during lockdown periods — when supply chain disruptions prevented effective price renegotiation — combined with compressed consumer demand in the institutional and food service channels, which account for a substantial proportion of Amul's B2B revenue.

4.2 Liquidity Analysis

Liquidity analysis evaluates GCMMF's capacity to meet short-term financial obligations as they fall due, a dimension of particular relevance given the organisation's extensive supply chain obligations to district cooperative unions, logistics service providers, and retail channel partners.

Fiscal Year	Current Assets (INR Crore)	Current Ratio	Quick Ratio
2018–19	8,120	1.72	1.18
2019–20	10,340	1.84	1.29
2020–21	9,860	1.68	1.14
2021–22	15,200	1.91	1.37
2022–23	18,440	2.03	1.46

Table 3: Liquidity Indicators – GCMMF (FY 2018–19 to FY 2022–23)

Current ratios consistently exceed the widely referenced benchmark threshold of 1.5 across all years examined, with the exception of the COVID-19 disruption year FY 2020 – 21 where the ratio contracted to 1.68 — still meaningfully above the minimum threshold. The sustained upward trend, culminating in a ratio of 2.03 in FY 2022–23, indicates progressive strengthening of the organisation's short-term financial buffer. This trajectory is consistent with GCMMF's post-pandemic strategic emphasis on working capital strengthening to mitigate future supply chain vulnerability — a lesson drawn from FY 2020–21 operational disruptions.

Quick ratios — which exclude inventory from current assets, thereby providing a more conservative liquidity assessment — follow a parallel pattern, ranging from 1.14 in the disruption year to 1.46 in FY 2022 – 23. The quick ratio approaching the 1.5 benchmark in the most recent year indicates that even under a scenario of complete inventory illiquidity, GCMMF maintains near-adequate capacity to discharge short-term obligations. Inventory composition warrants note in this context: a substantial proportion of Amul's inventory comprises perishable dairy products with rapid turnover characteristics, making the inventory exclusion in the quick ratio calculation more conservative than it would be for manufacturing enterprises with longer inventory holding periods.

4.3 Solvency Analysis

Long-term solvency analysis examines GCMMF's capacity to meet obligations extending beyond the immediate operating cycle, with particular focus on the organisation's capital structure and debt burden relative to equity and earnings.

Fiscal Year	Total Debt (INR Crore)	Debt-to-Equity Ratio	Interest Coverage Ratio
2018–19	1,240	0.23	14.2
2019–20	1,410	0.20	16.8
2020–21	1,680	0.24	12.3
2021–22	1,320	0.16	21.4
2022–23	1,150	0.12	26.7

Table 4: Solvency Indicators – GCMMF (FY 2018 – 19 to FY 2022 – 23)

The debt-to-equity ratio data reveal a particularly striking pattern: a consistent and meaningful decline from 0.23 in FY 2018–19 to 0.12 in FY 2022–23. This trajectory indicates that GCMMF has been progressively deleveraging its capital structure — reducing total debt while simultaneously growing its equity base through accumulated reserves — a financial management posture typically associated with conservative, low-risk institutional philosophies. For cooperative enterprises, this conservatism is strategically rational: cooperative members bear equity risk without the liquidity mechanisms (stock markets, secondary trading) available to investor-owned firm shareholders, making low leverage a governance-aligned rather than merely financially-motivated choice (Chaddad & Cook, 2004).

Interest coverage ratios demonstrate a pronounced and consistent upward movement, from 14.2x in FY 2018–19 to 26.7x in FY 2022–23, with the expected dip to 12.3x during the COVID-19 affected year. Even at the cyclical low, an interest coverage ratio exceeding 12x indicates substantial earnings headroom above debt service obligations — GCMMF's earnings before interest and tax were more than twelve times its interest expense even in its worst recent year. The most recent ratio of 26.7x places GCMMF in a position of exceptional debt service comfort, substantially in excess of the 3x minimum conventionally considered adequate for investment-grade commercial enterprises. This position significantly de-risks GCMMF's balance sheet against future revenue shocks.

4.4 Synthesis: The Cooperative Financial Performance Profile

The consolidated findings across profitability, liquidity, and solvency dimensions delineate a coherent and internally consistent financial performance profile. GCMMF's financial architecture reflects deliberate design choices aligned with cooperative enterprise logic rather than conventional commercial optimisation. Thin but stable profit margins coexist with strong revenue growth, conservative leverage, and progressively strengthening liquidity — a combination that prioritises institutional durability and farmer welfare over shareholder value maximisation. This profile can be characterised as a 'sustainability-first' financial strategy wherein the organisation deliberately operates at the boundary between commercial viability and social mission delivery.

The COVID-19 period (FY 2020–21) constitutes the most significant stress event in the study period and provides the most diagnostic evidence of GCMMF's financial resilience. Across all three analytical categories, ratios declined but remained above critical thresholds — profitability contracted but the organization remained profitable; liquidity ratios declined but remained positive; interest coverage fell but stayed at multiples far above minimum thresholds. This multi-dimensional resilience under simultaneous demand, supply chain, and macroeconomic stress validates the structural soundness of GCMMF's conservative financial architecture.

2. Conclusion & Future Scope

Amul (GCMMF) has demonstrated impressive financial performance, with an 8% revenue growth in FY 24, reaching ₹ 59,445 crore. The group's turnover hit ₹ 80,000 crore in FY 24, up from ₹ 72,000 crore in FY 23. Key drivers include product diversification, innovation, and an expanded distribution network.

5.1 Financial Highlights:

- i. Revenue Growth: 8% in FY24, reaching ₹ 59,445 crore
- ii. Group Turnover: ₹ 80,000 crore in FY 24
- iii. Profitability: Strong consumer demand and volume growth
- iv. Liquidity: Robust distribution network and efficient logistics
- v. Solvency: Strong brand presence and quality products

5.2 Future Scope:

- i. Expansion in to new markets and product categories
- ii. Increased focus on sustainability and CSR initiatives
- iii. Continued innovation and product launches
- iv. Export growth, particularly in the US and other international markets

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